

Matgeo-2.2.11

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Question

The plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1} \alpha$ with the x -axis. Find the value of α .

Solution

$$\vec{n} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix}, \quad (\text{normal vector}) \quad (1)$$

$$\vec{a} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad (\text{direction vector of x-axis}) \quad (2)$$

$$\cos \phi = \frac{\vec{n} \cdot \vec{a}}{\|\mathbf{n}\| \|\mathbf{a}\|} \quad (3)$$

$$= \frac{2}{\sqrt{2^2 + (-3)^2 + 6^2}} \quad (4)$$

$$= \frac{2}{7} \quad (5)$$

Solution(Continuation)

$$\theta = 90^\circ - \phi \quad (6)$$

$$\sin \theta = \cos \phi = \frac{2}{7} \quad (7)$$

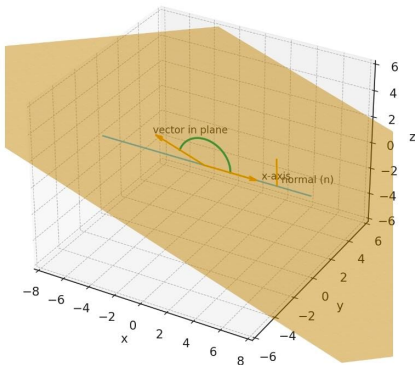
$$\text{So, } \theta = \sin^{-1} \left(\frac{2}{7} \right)$$

Conclusion

Conclusion: The plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}\left(\frac{2}{7}\right)$ with the x -axis. Hence, the required value of α is $\alpha = \frac{2}{7}$.

Graphical Representation

Plane $2x - 3y + 6z - 11 = 0$, x-axis, a vector in the plane and the normal
(arc shows angle between x-axis and the plane)



Figure